

Quotient Laws of Exponents*Question Bank - Additional 3.02*

Show your work. Circle your final answer.

For questions 1 through 3, show the quotient of powers law, then evaluate.

1. $\frac{5^{10}}{5^7}$

2. $\frac{7^5}{7^4}$

3. $\frac{(-4)^4}{(-4)^2}$

4. When raising a fraction to a power, the numerator and the denominator are each raised to the _____ power.

For questions 5 and 6, apply the power of a quotient law. Do not simplify the fraction.

5. $\left(\frac{-3}{-6}\right)^6$

6. $\left(\frac{7}{2}\right)^4$

The zero exponent.

7. Fill in the missing value in the pattern: $5^4 = 625$, $5^3 = 125$, $5^2 = 25$, $5^1 = 5$,
 $5^0 =$ _____

8. Evaluate 12^0 and $\left(\frac{8}{9}\right)^0$.

For questions 9 and 10, rewrite the expression using a law of exponents, name the law, then evaluate.

9. $\frac{3.2^3}{3.2^1}$

10. $\left(\frac{1.5}{0.5}\right)^3$

Working backwards.

11. Find the value of n that makes each statement true.

a. $\frac{2^n}{2^4} = 2^5$

b. $\left(\frac{2}{3}\right)^n = \frac{16}{81}$

c. $\frac{9^6}{9^n} = 1$

12. Which expression is equivalent to $\frac{6^{10}}{6^5}$?

- ☐ A. 6^5
- ☐ B. 6^2
- ☐ C. 1^5
- ☐ D. 6^{15}

13. Select ALL of the expressions whose value is 1.

☐ A. 8^0

☐ B. $\frac{2^5}{2^5}$

☐ C. $\left(\frac{3}{7}\right)^0$

☐ D. $\frac{7^2}{49}$

☐ E. $\frac{2^5}{2^4}$

☐ F. 0^5